

Bank Account Verification & Name Fetching APIs

Complete Guide on Penny Drop, Penniless Verification & Provider Integration (India)

Prepared for MK-GOLD ERP & Billing System | September 2026

1. Executive Overview & The "Free API" Misconception

A common question during fintech and gold-buying software development is whether a free public API (like <https://ifsc.razorpay.com/>) can verify a customer's bank account number and fetch the account holder's name. The answer is strictly NO.

- **IFSC Lookup API (Free):** Razorpay and others provide free IFSC lookups, but IFSC codes only identify the bank branch, city, and state. They have no access to customer account data.
- **Account & Name Lookup (Restricted):** Under Reserve Bank of India (RBI) and National Payments Corporation of India (NPCI) privacy guidelines, bank account balances and account holder identities are strictly protected banking records.
- **How Real Verification Works:** Querying an account requires using licensed Bank Account Validation (BAV) APIs via authorized payment aggregators (Razorpay, Cashfree, Decentro, Setu) using Penny Drop or NPCI BAV.

2. The Three Methods of Bank Account Verification

- **1. Penny Drop (IMPS Micro-Deposit):** The API sends Rs 1 (or Rs 0.01) to the customer's account via IMPS. The destination bank responds with the UTR and the registered beneficiary name (e.g., "BHARATH KUMAR"). Takes 1-3 seconds.
- **2. Penny-Less / Penniless Verification:** A direct inquiry protocol via NPCI that queries the account status without actually crediting money. Faster and avoids micro-refund accounting, supported by major banks.
- **3. Reverse Penny Drop (UPI):** The merchant displays a Rs 1 dynamic QR / UPI intent. The customer pays from their UPI app. NPCI webhook instantly returns the verified account holder name and linked bank account.

3. Comparative Summary of Leading Indian Providers

Provider	Method	Speed	Name Match	Avg Cost / Check	Best For
RazorpayX	Penny Drop + Reverse	1-2 sec	Yes (0-100 Score)	Rs 1.50 - 2.50	Easiest setup & UI
Cashfree BAV	Penny Drop + Penniless	Sub-second	Yes (Fuzzy Match)	Rs 1.20 - 2.00	High volume & speed
Setu (Pine Labs)	Penny Drop + UPI	1-2 sec	Yes (AI match)	Rs 1.50 - 2.20	Enterprise KYC suite
Decentro	Penny Drop + Penniless	1-2 sec	Yes (Score)	Custom / Tiered	Full banking SDKs
Surepass.io	Penny Drop + Penniless	1-3 sec	Yes (Verified)	Rs 1.00 - 1.80	NBFCs & Gold loans

Detailed Provider Technical Specifications

API Endpoints, Request/Response Payloads & Integration Details

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Provider 1: RazorpayX Bank Account Validation API

RazorpayX provides instant fund account validation. Once enabled on the Razorpay Dashboard, you send the bank account number and IFSC code.

API Endpoint & Request:

```
POST https://api.razorpay.com/v1/fund_accounts/validations
Headers:
  Authorization: Basic base64(YOUR_KEY_ID:YOUR_KEY_SECRET)
  Content-Type: application/json

Payload:
{
  "account_number": "409000392345", // Your RazorpayX payout virtual account
  "fund_account": {
    "account_type": "bank_account",
    "bank_account": {
      "name": "Bharath Kumar",
      "account_number": "10710100283243",
      "ifsc": "CIUB0000107"
    }
  },
  "amount": 100, // 100 paise = INR 1.00
  "currency": "INR"
}
```

API Response:

```
{
  "id": "fav_J92Ksl01kd92La",
  "status": "completed",
  "validation_results": {
    "account_status": "active",
    "registered_name": "BHARATH KUMAR",
    "name_match_score": 100
  },
  "utr": "425118920192"
}
```

Provider 2: Cashfree Bank Account Verification (BAV)

Cashfree is known for having one of the lowest latency verification engines in India. Supports synchronous (instant) verification.

API Endpoint & Request:

```
POST https://api.cashfree.com/verification/bank-account/sync
```

Headers:

```
x-client-id: <YOUR_APP_ID>
x-client-secret: <YOUR_SECRET_KEY>
Content-Type: application/json
```

Payload:

```
{
  "name": "Bharath Kumar",
  "phone": "9876543210",
  "bank_account": "10710100283243",
  "ifsc": "CIUB0000107"
}
```

API Response:

```
{
  "status": "SUCCESS",
  "subCode": "200",
  "account_status": "VALID",
  "account_exists": "YES",
  "name_at_bank": "BHARATH KUMAR",
  "name_match_score": "1.0",
  "utr": "319201948201"
}
```

Architecture & Implementation in MK-GOLD

Secure Backend Proxy, Auto-Population in React & Fraud Protection

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Recommended Architecture for MK-GOLD ERP

SECURITY RULE: Never expose payment gateway API keys in frontend React code. Always build a secure backend proxy endpoint in the Express server to handle verification.

1. Architecture Flow:

- **Step 1 (Branch Staff):** Branch staff enters customer Account Number and IFSC in the Customer Bank popup.
- **Step 2 (Frontend Call):** React calls the internal backend endpoint: POST /api/branch/customer-bank/verify-account.
- **Step 3 (Backend Execution):** Express backend securely calls RazorpayX or Cashfree with server-side credentials.
- **Step 4 (Auto-Population):** Backend returns registered_name. Frontend auto-populates "Account Holder Name" and sets it to read-only, preventing clerical errors.

2. Backend Express Controller Snippet (Node.js):

```
// backend/controllers/branch/customer-bank.js
const axios = require('axios');

async function verifyBankAccount(req, res) {
  const { accountNo, ifscCode, declaredName } = req.body;
  if (!accountNo || !ifscCode) {
    return res.status(400).json({ status: false, message: 'Account number and IFSC are required' });
  }

  try {
    // Calling Cashfree Bank Account Verification API
    const cfResponse = await axios.post(
      'https://api.cashfree.com/verification/bank-account/sync',
      { bank_account: accountNo, ifsc: ifscCode, name: declaredName || '' },
      {
        headers: {
          'x-client-id': process.env.CASHFREE_APP_ID,
          'x-client-secret': process.env.CASHFREE_SECRET_KEY,
        }
      }
    );

    const data = cfResponse.data;
    if (data.status === 'SUCCESS' && data.account_status === 'VALID') {
      return res.json({
        status: true,
        data: {
          accountHolderName: data.name_at_bank,
          accountStatus: data.account_status,
          utr: data.utr,
          nameMatchScore: data.name_match_score,
        }
      });
    } else {
      return res.json({ status: false, message: 'Invalid or inactive bank account' });
    }
  } catch (err) {
    console.error('Bank verification error:', err.response?.data || err.message);
    return res.status(500).json({ status: false, message: 'Verification service error' });
  }
}
```

3. Summary Checklist to Go Live:

- **1. Sign Up:** Create an account on Razorpay (RazorpayX) or Cashfree (Verification Suite).
- **2. Complete KYC:** Submit business documents (Company PAN, GST, Board Resolution) for BAV API access.
- **3. Load Wallet:** Maintain a small prepaid balance (e.g. Rs 500 - Rs 1,000) to cover micro-deposits and verification fees.
- **4. Add Env Keys:** Add CASHFREE_APP_ID & CASHFREE_SECRET_KEY into backend/.env.

